5(a)	volt / ampere	**B1**
5(b)	$R = \rho L / A$	**C1**
	$L = (1.8 \times 0.38 \times 10^{-6}) / 9.6 \times 10^{-7}$	**C1**
	$= 0.71$ m	**A1**
5(c)(i)	thermal energy is dissipated in resistor Y	**B1**
5(c)(ii)	$V / 1.2 = 1.8 / (1.8 + 0.6)$	**C1**
	$V = 0.90$ V	**A1**
	or	
	$I = 1.2 / (1.8 + 0.6)$ (= 0.50)	**(C1)**
	$V = 0.50 \times 1.8$ $= 0.90$ V	**(A1)**
5(d)(i)	remain the same	**B1**
5(d)(ii)	decrease	**B1**
5(e)(i)	$1 / R = 1 / 1.8 + 1 / 3.6$ $R = 1.2\,\Omega$	**A1**

Question	Answer	Marks
5(e)(ii)	I = 1.2 / (1.2 + 0.60)	**C1**
	= 0.67 A	**A1**
	or	
	V_Y = 1.2 × 0.60 / (1.2 + 0.60) (= 0.40)	**(C1)**
	I = 0.40 / 0.60 = 0.67 A	**(A1)**

Question	Answer	Marks